Longest Common Substring

Problem Statement

Given two strings S1 and S2. Find the longest common substrings between S1 and S2.

Example:-

S1 = "LCLC"

S2 = "CLCL"

S1 = "LCLC"

S2 = "CLCL"

S1 = "LCLC"

S2 = "CLCL"

Solution:

- Compare all substrings?- $O(2^n)$
- Better way?

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S1 = “LCLC”

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S1 = “LCLC”

S2 = “CLCL”

S1 = “LCLC”

S2 = “CLCL”

Max:0, Result: {}

	L	C	L	C
C	0	0	0	0
L	0	0	0	0
C	0	0	0	0
L	0	0	0	0

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Max:0, Result: {}

	L	C	L	C
C	0	0	0	0
L	0	0	0	0
C	0	0	0	0
L	0	0	0	0

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Example:-

S1 = “LCLC”

S2 = “CLCL”

S1 = “LCLC”

S2 = “CLCL”

S1 = “LCLC”

S2 = “CLCL”

Max:1, Result: {C}

	L	C	L	C
C	0	1	0	0
L	0	0	0	0
C	0	0	0	0
L	0	0	0	0

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S1 = “LCLC”

S2 = “CLCL”

S1 = “LCLC”

S2 = “CLCL”

Max:1, Result: {C}

	L	C	L	C
C	0	1	0	0
L	0	0	0	0
C	0	0	0	0
L	0	0	0	0

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S1 = “LCLC”

S2 = “CLCL”

S1 = “LCLC”

S2 = “CLCL”

Max:1, Result: {C}

	L	C	L	C
C	0	1	0	1
L	0	0	0	0
C	0	0	0	0
L	0	0	0	0

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Example:-

S1 = “LCLC”

S2 = “CLCL”

S1 = “LCLC”

S2 = “CLCL”

S1 = “LCLC”

S2 = “CLCL”

Max:1, Result: {C, L}

	L	C	L	C
C	0	1	0	1
L	1	0	0	0
C	0	0	0	0
L	0	0	0	0

Problem Statement

Given two strings S1 and S2. Find the longest common substrings between S1 and S2.

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S2 = "CLCL"

S1 = "LCLC"

S2 = "CLCL"

S1 = "LCLC"

S2 = "CLCL"

Max:1, Result: {C, L}

	L	C	L	C
C	0	1	0	1
L	1	0	0	0
C	0	0	0	0
L	0	0	0	0

Problem Statement

Given two strings S1 and S2. Find the longest common substrings between S1 and S2.

Example:-

S1 = “LCLC”

S2 = “CLCL”

S1 = “LCLC”

S2 = “CLCL”

S1 = “LCLC”

S2 = “CLCL”

Max:2, Result: {CL}

	L	C	L	C
C	0	1	0	1
L	1	0	2	0
C	0	0	0	0
L	0	0	0	0

Problem Statement

Given two strings S1 and S2. Find the longest common substrings between S1 and S2.

Example:-

S1 = “LCLC”

S2 = “CLCL”

S1 = “LCLC”

S2 = “CLCL”

S1 = “LCLC”

S2 = “CLCL”

Max:2, Result: {CL,LC}

	L	C	L	C
C	0	1	0	1
L	1	0	2	0
C	0	2	0	0
L	0	0	0	0

Problem Statement

Given two strings S1 and S2. Find the longest common substrings between S1 and S2.

Example:-

S1 = “LCLC”

S2 = “CLCL”

S1 = “LCLC”

S2 = “CLCL”

S1 = “LCLC”

S2 = “CLCL”

Max:3, Result: {CLC}

	L	C	L	C
C	0	1	0	1
L	1	0	2	0
C	0	2	0	3
L	0	0	0	0

Problem Statement

Given two strings S1 and S2. Find the longest common substrings between S1 and S2.

Example:-

S1 = “LCLC”

S2 = “CLCL”

S1 = “LCLC”

S2 = “CLCL”

S1 = “LCLC”

S2 = “CLCL”

Max:3, Result: {CLC}

	L	C	L	C
C	0	1	0	1
L	1	0	2	0
C	0	2	0	3
L	1	0	0	0

Problem Statement

Given two strings S1 and S2. Find the longest common substrings between S1 and S2.

Example:-

S1 = “LCLC”

S2 = “CLCL”

S1 = “LCLC”

S2 = “CLCL”

S1 = “LCLC”

S2 = “CLCL”

Max:3, Result: {CLC, LCL}

	L	C	L	C
C	0	1	0	1
L	1	0	2	0
C	0	2	0	3
L	1	0	3	0

Problem Statement

Given two strings S1 and S2. Find the longest common substrings between S1 and S2.

Example:-

S1 = “LCLC”

S2 = “CLCL”

S1 = “LCLC”

S2 = “CLCL”

S1 = “LCLC”

S2 = “CLCL”

Max:3, Result: {CLC, LCL}

	L	C	L	C
C	0	1	0	1
L	1	0	2	0
C	0	2	0	3
L	1	0	3	0

Recurrence Relation

$$DP[i][j] = \begin{cases} 1 + DP[i-1][j-1] & \text{iff } s1[i] == s2[j] \\ 0 & \text{iff } s1[i] != s2[j] \end{cases}$$

```
vector<string> longestCommonSubstring(string s1, string s2){
    int DP[s1.length()][s2.length()];
    vector<string> ans;
    int maxLen = INT_MIN;

    for(int i=0; i<s1.length(); i++){
        for(int j=0; j<s2.length(); j++){
            if(s1[i] == s2[j]){
                //if the match happened in the first row or column then longest common substring is 1.
                if(i==0 || j==0)
                    DP[i][j] = 1;
                else{
                    //recur for DP[i-1][j-1]
                    DP[i][j] = DP[i-1][j-1] + 1;

                    //if this new common string is longer than the longest found so far
                    if(DP[i][j] > maxLen){
                        maxLen = DP[i][j];
                        ans.clear();
                        ans.push_back(s1.substr(i-maxLen+1, maxLen));
                    }
                    else if(DP[i][j] == maxLen){
                        ans.push_back(s1.substr(i-maxLen+1, maxLen));
                    }
                }
            }
            else
                DP[i][j]=0;
        }
    }
    return ans;
}
```